Verification of whether the final matrix is in rref

1. Row 1 and 2 are non-zero rows. Row 3 is a zero row. ∴ The condition that all the non-zero rows must be above the zero rows is True.

2. Row 1 Leading entry = 1 in Column 1
   Row 2 Leading entry = 1 in Column 2
   ∴ The condition that for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it is True.

3. Row 2 Column 1 = Row 3 Column 1 = 0
   Row 3 Column 2 = 0
   ∴ The condition that for each column, having a leading entry, all positions below it must contain 0 is True.

4. The condition that each leading entry must be 1 is True.

5. Row 1 Column 2 = 0
   ∴ The condition that for each column having a leading entry, all positions above it must contain 0 is True.

The matrix $\begin{bmatrix} 1 & 0 & 5 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$ is in rref □

- There are no rows in the matrix of the form $[0\ 0 \ldots\ldots 0\ b]$ where $b \neq 0$. By Theorem 2, the corresponding system of equations are consistent.
- All the columns of A are pivot columns. ∴ By defn, there are no free variables.
- By Theorem 2, the solution is unique.

∴ $x_1 = 5,\ x_2 = 3.$ $\boxed{x = \begin{bmatrix} 5 \\ 3 \end{bmatrix}}$ □

In Exercises 23 and 24, mark each statement True or False. Justify each answer.

23)a) A linear transformation $T: \mathbb{R}^n \to \mathbb{R}^m$ is completely determined by its effect on the columns of the $n \times n$ identity matrix.